Product Development and Design

Week 6 Conjoint Analysis and Product Architecture

MKT3717 Product and Brand Management

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Agenda

- So far: NPD fuzzy front end
  - Opportunity Identification, Concept Generation, Concept Testing
- From now: Product development and launch
  - Conjoint analysis as concept and market testing tool (continued next week)
  - Product architecture and design
  - Product launch strategy (article discussion)
- Next week: Market experiment design and project management

5. Product Design and Launch

Motivation for Conjoint Analysis

- In product development, we often want to understand how people value different **attributes** of a product.

- For developing a new salsa product for young consumers, which are the important attributes?
  - Thickness of salsa
  - Spiciness
  - Color of salsa

- For a hotel in Sentosa, how much do consumers value?
  - Slightly larger room
  - Ocean view
  - Price

Trade-off (Conjoint) Analysis

- How people value different **attributes** of a product?
  - Key assumption: Product is a bundles of attributes and customers combine values from attributes to generate values of product

- Can be used in **concept evaluation**
  1. Put the determinant attributes together in combinations or sets.
  2. Respondents rank among these sets in order of preference or make choices.
  3. Respondents “trade off” what is less desirable for what they desire.
  4. Optimal levels of each attribute is sought.

Terminology

- Attributes
  - Important product characteristics
- Levels
  - Quantities or qualities of attributes
- Utility
  - Value that customers place on each level

https://conjointly.com/guides/what-is-conjoint-analysis/

Conjoint Analysis

: Survey-based technique used to determine how people value different attributes of a product/service

- **Approach**:

1. Each subject sees many products with different combinations of attributes and is asked to rate their preference for the products
   - In other iterations of conjoint analysis, people can be asked to make choices instead of rating preferences
2. We use regression analysis to analyze the reported preferences to infer the valuation of each attribute

Conjoint Analysis: Salsa Example

- 3 attributes of salsa
  - Thickness (3 levels)
    - Regular, thick, extra-thick
  - Spiciness (3 levels)
    - Mild, medium-hot, extra-hot
  - Color (2 levels)
    - Green, red
- 3 x 3 x 2 = 18 possible combinations

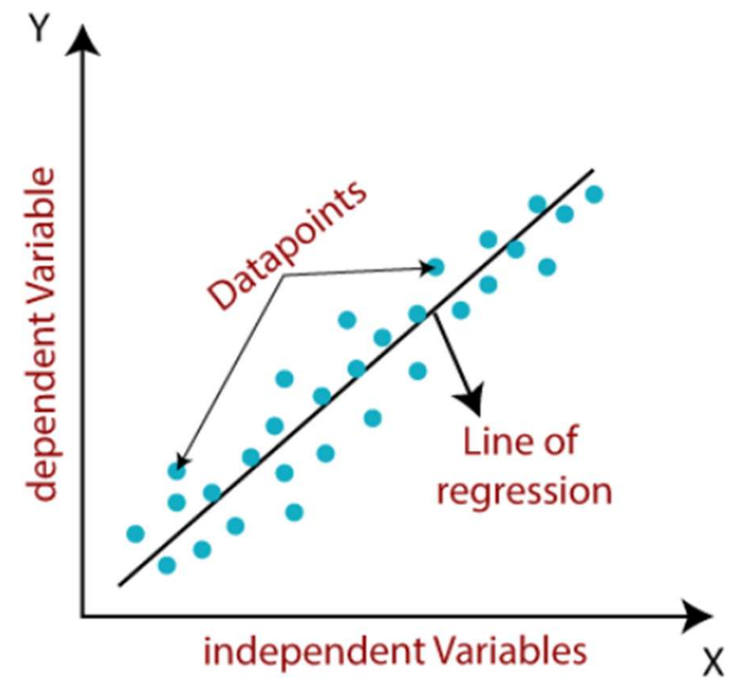
Thickness	Spiciness	Color	Actual Ranking*
Regular	Mild	Red	4
Regular	Mild	Green	3
Regular	Medium-Hot	Red	10
Regular	Medium-Hot	Green	6
Regular	Extra-Hot	Red	15
Regular	Extra-Hot	Green	16
Thick	Mild	Red	2
Thick	Mild	Green	1
Thick	Medium-Hot	Red	8
Thick	Medium-Hot	Green	5
Thick	Extra-Hot	Red	13
Thick	Extra-Hot	Green	11
Extra-Thick	Mild	Red	7
Extra-Thick	Mild	Green	9
Extra-Thick	Medium-Hot	Red	14
Extra-Thick	Medium-Hot	Green	12
Extra-Thick	Extra-Hot	Red	17
Extra-Thick	Extra-Hot	Green	18

* 1 = most preferred, 18 = least preferred.

Quick Look: Regression analysis

$$Y = a + b\,X$$

- **b** is the coefficient of interest that shows the relationship between X and Y
  - Magnitude of b
    - *One unit increase in X is associated with b units increase in Y*
  - Statistical significance of b
    - *Results will show p-value*
    - *We say the relationship is statistically significant if p-value is sufficiently small (typically $p<0.05$)*



Conjoint Analysis

- Once the data is in, we simply estimate the following regression:

$$Rank_i = \beta_0 + \beta_1\ Thick_i + \beta_2\ ExtraThick_i + \beta_3\ MediumHot_i + \beta_4\ ExtraHot_i + \beta_5\ Red_i + e_i$$

- Note:

Thickness	Spicy Level	Color
• Regular (baseline) • Thick • Extra Thick	• Mild (baseline) • Medium Hot • Extra Hot	• Green (baseline) • Red

You can download Excel Analysis Toolpak:
https://support.microsoft.com/en-us/office/load-the-analysis-toolpak-in-excel-6a63e598-cd6d-42e3-9317-6b40ba1a66b4

Interpretation of Results

- You can use estimated coefficients to make inference on importance of each attribute!

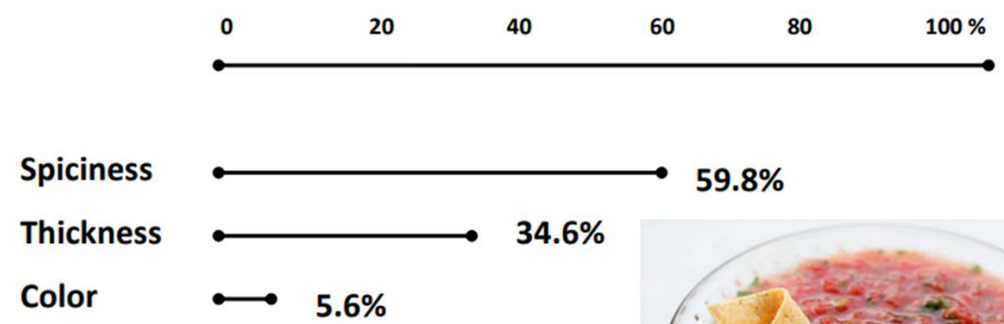
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.977736
R Square	0.955968
Adjusted R Squa	0.937622
Standard Error	1.333333
Observations	18

ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regression	5	463.1667	92.63333	52.10625	1.01E-07
Residual	12	21.33333	1.777778		
Total	17	484.5			

	Coefficients	*andard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower*	*pper*
Intercept	3.333333	0.7698	4.330127	0.000978	1.656082	5.010584	1.656082	5.010584
Thick	-2.33333	0.7698	-3.03109	0.010445	-4.01058	-0.65608	-4.01058	-0.65608
Extra-thick	3.833333	0.7698	4.979646	0.00032	2.156082	5.510584	2.156082	5.510584
Med-hot	4.833333	0.7698	6.278684	4.08E-05	3.156082	6.510584	3.156082	6.510584
Extra-hot	10.66667	0.7698	13.85641	9.59E-09	8.989416	12.34392	8.989416	12.34392
Red	1	0.628539	1.59099	0.137597	-0.36947	2.36947	-0.36947	2.36947



Relative importance of attributes in Salsa Example
(Textbook p. 148)

Conjoint analysis (2)

- Now, let's consider designing hotel staycation package in Sentosa!

- For simplicity, suppose there are only three attributes:

Room size	View	Price
• $40m^2$ • $50m^2$	• Garden view • City view • Ocean view	• $300 per night • $350 per night

- There are 2 x 3 x 2 = 12 possible combinations of rooms

Conjoint analysis (2)

- Now, we conduct a survey where each subject is asked to rate each of the 12 versions of the hotel room with a score from 0 to 10, where 10 represents highest degree of preference.

- Notes:
  - We can randomize the order to minimize order effects

Conjoint analysis (2)

- Once the data is in, we simply estimate the following regression:

$$Y_i = \beta_0 + \beta_1 \text{ LargeRoom}_i + \beta_2 \text{ CityView}_i + \beta_3 \text{ OceanView}_i + \beta_4 \text{ HighPrice}_i + e_i$$

- Note:

Room size	View	Price
• $40m^2$ • $50m^2$	• Garden view • City view • Ocean view	• $300 per night • $350 per night

Interpreting results

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.98169363							
R Square	0.963722382							
Adjusted R Square	0.942992315							
Standard Error	0.673565495							
Observations	12							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	4	84.36666667	21.09166667	46.48911047	3.97666E-05			
Residual	7	3.175833333	0.453690476					
Total	11	87.5425						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	2.083333333	0.434784658	4.791644087	0.001985359	1.055230988	3.111435679	1.055230988	3.111435679
LargeRoom (+10m2)	1.416666667	0.38888322	3.642910247	0.008255163	0.497103974	2.33622936	0.497103974	2.33622936
CityView	2.05	0.476282729	4.304166149	0.003548409	0.923770308	3.176229692	0.923770308	3.176229692
OceanView	5.825	0.476282729	12.23013064	5.5967E-06	4.698770308	6.951229692	4.698770308	6.951229692
HighPrice (+$50)	-1.683333333	0.38888322	-4.328634529	0.003443595	-2.602896026	-0.76377064	-2.602896026	-0.76377064

Interpreting results (2)

- The estimated model is:

$$Y_i = 2.08 + 1.42\ \text{LargeRoom}_i + 2.05\ \text{CityView}_i + 5.83\ \text{OceanView}_i - 1.68\ \text{HighPrice}_i$$

- We can use the estimated coefficients to estimate the value of each attribute:
  - Increasing price from \$300 to \$350 decreases preference rating by 1.68
  - Increasing room size from $40m^2$ to $50m^2$ increases preference rating by 1.42
    - This is 1.42/1.68 = 85% of the magnitude of \$50 price increase
    - We can approximate that consumer would be indifferent between $40m^2$ room at \$300 and $50m^2$ room at \$342
  - Changing view from garden to city increases preference rating by 2.05
    - …indifferent between garden view room at \$300 and city view room at \$361
  - Changing view from garden to ocean increases preference rating by 5.83
    - …indifferent between garden view room at \$300 and ocean view room at \$474

Class Exercise

- Average ranking for 1.5kg =
- Average ranking for 2kg =
- Average difference in Weight?

- Average ranking for 4hr =
- Average ranking for 6hr =
- Average difference in Battery life?

- Average ranking for HP =
- Average ranking for Dell =
- Average difference in Brand?

What is the most/least important attribute for this consumer?

Three attributes of notebook computers:	
Weight (1.5kg or 2kg)	
Battery life (4 hours or 6 hours)	
Brand (HP, Dell)	
Combination	Rank
1.5kg, 4 hours, HP	4
2kg, 6 hours, Dell	5
2kg, 4 hours, HP	8
1.5kg, 6 hours, HP	3
1.5kg, 4 hours, Dell	2
2kg, 6 hours, HP	7
2kg, 4 hours, Dell	6
1.5kg, 6 hours, Dell	1

Notes on Conjoint Analysis

- You can estimate the parameters at individual level or estimate the parameters across all subjects.

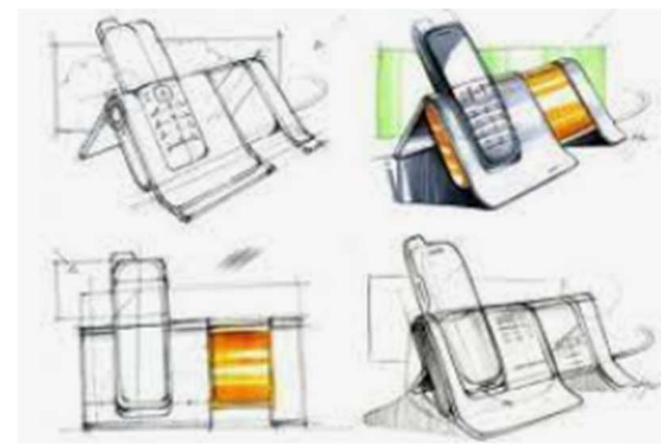
- Our simple example involved subjects evaluating all possible combinations of the attributes. This is called a **full factorial design**.
  - There are ways to efficiently show only a subset of possible combinations, and this approach is called **fractional factorial design.**

- You can be flexible with the dependent variable
  - E.g., you may ask subjects to evaluate in any scoring system you would like; rank-order, rating, choice between pairs (Choice-based), or hybrid (adaptive)

Use of Conjoint Analysis

Why? When?

- Product concept testing
  - Caveat: possibly less useful for new-to-the-world product
- Industrial product innovation (work well!)
  - Identifying which set of attributes most preferred by customers
- Most desired needs for specific segment of customers
- Brand and price trade-off (to assess brand value)



When to Use Conjoint?

- Product value can be approximated from the sum of the parts
  - Computers, financial products, pack & size combination
  - No: soft drinks, clothing design, life partner
- Attributes do not interact
  - Credit card features, cell phone packages, car rental options
  - No: Fragrances, meal ingredients, speeches
- Attributes can be described in discrete and unambiguous terms
  - Apple iOS vs. Android, 2 vs 4 door, $100 vs $200, glass vs plastic
  - Friendly, clean, quick, sexy, sporty, ambitious

For deeper understanding, **read Crawford Chapter 6. p.143-151, Chapter 7. 180-192**

Group Exercise

You work for Tinder. Your task is to use conjoint analysis to determine what attributes in a profile lead to successful matches. Eventually, you want to use insights gained from this analysis to develop a premium service that provides personalized recommendations to customers on how to setup the profile to maximize the chances of a successful match. Can you brainstorm how you would tackle this problem?

Hint: The goal here is to maximize the chances of your client getting 'swiped right'

The great thing is, the data on customers' preferences is already there!

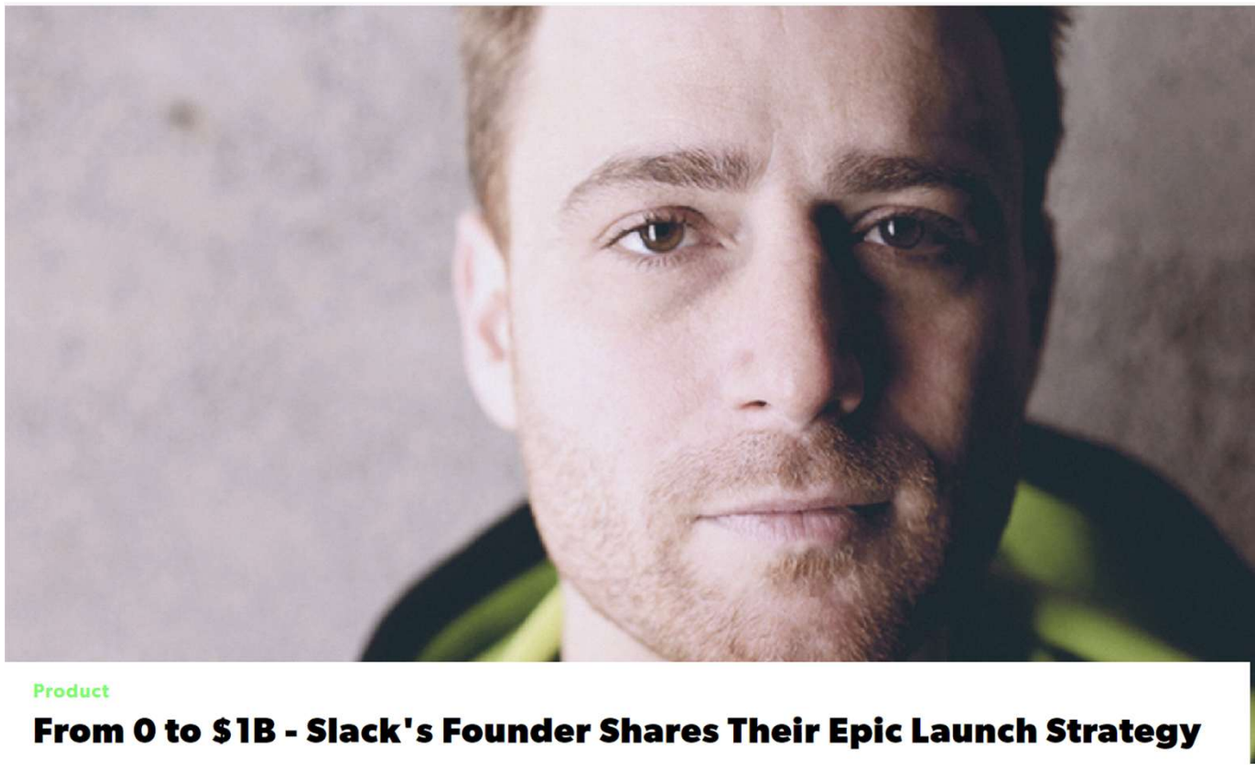
Limitations

- Must be able to break the product down into discrete attributes and customers must be able to combine the features rationally when evaluating the products (product value = sum of attributes, with no interaction)
- Complex products may have too many combinations of attributes; subjects may get frustrated/fatigued and resort to simplification strategies
- Subjects may be forced to think about attributes that they have never thought about (or do not care), which could lead to unreliable responses
- Hard to be properly incentivized

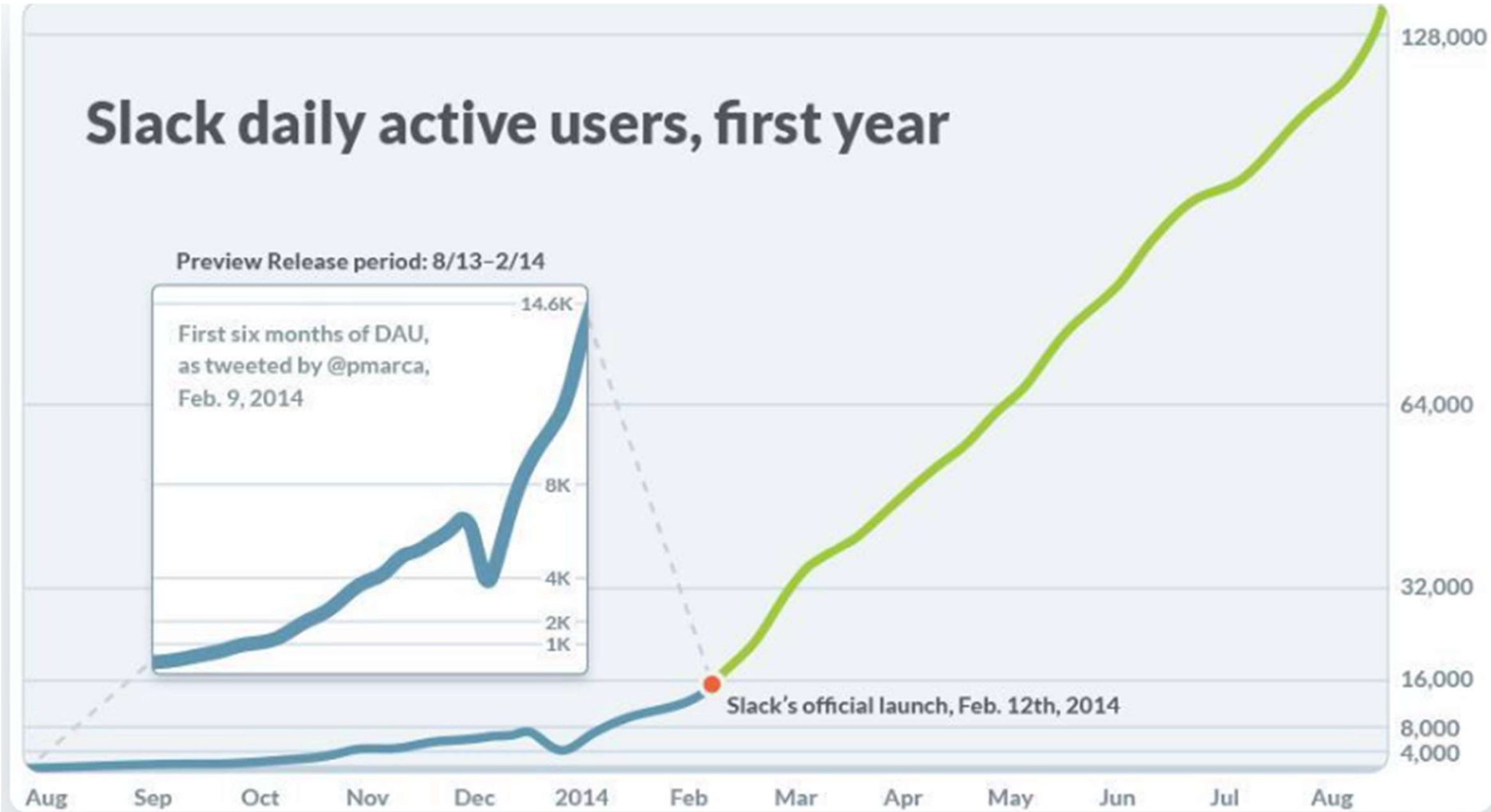
As long as you understand its limitations, conjoint analysis can be an extremely useful and versatile tool in your arsenal!

Article Discussion on Product Launch

What are the key takeaways behind Slack's product development and launch strategy?

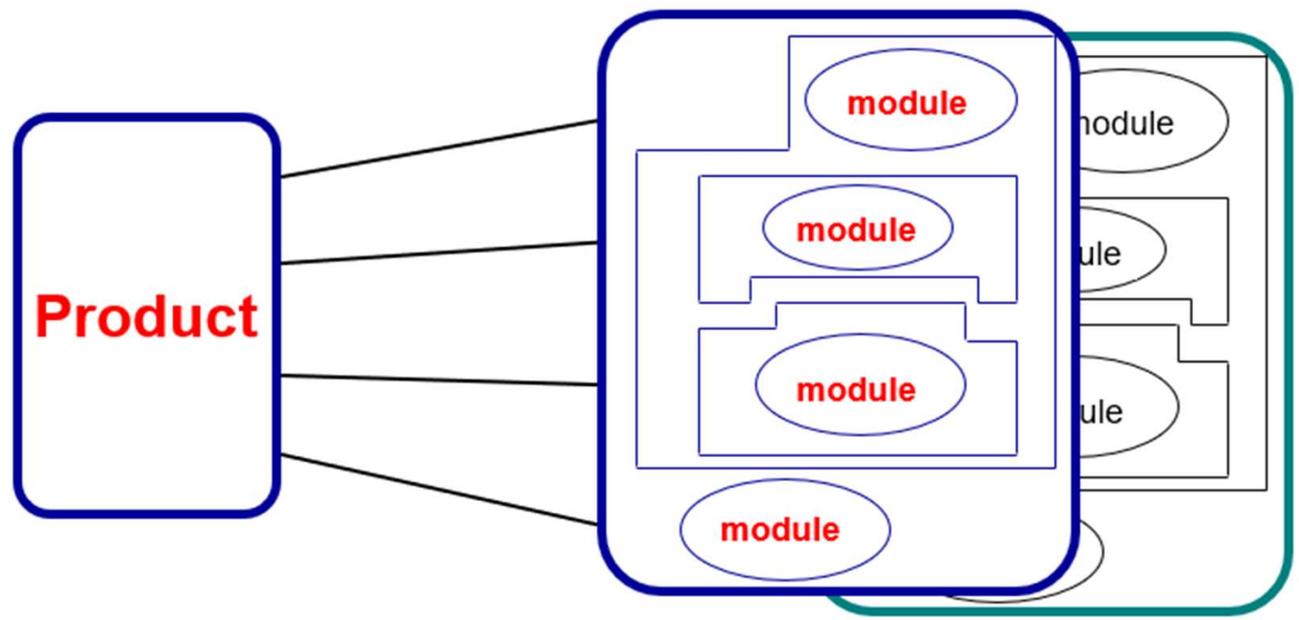


Slack's Product Development and Launching



Product Architecture

- The arrangement of functional elements into chunks which become the building blocks for the product or family of products



Product Architecture Considerations

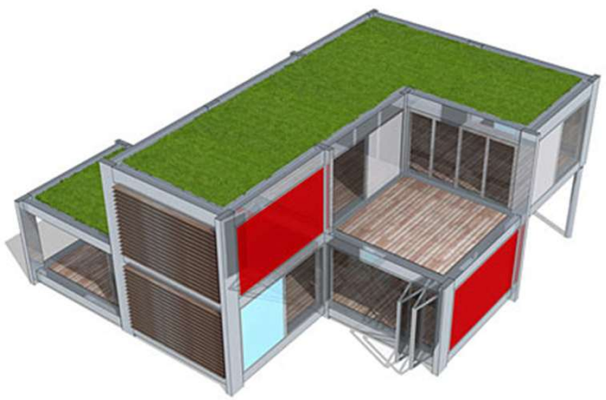
Roche Bobois

Sectional Furniture

DIN Audio System

Alpine

Lovetann



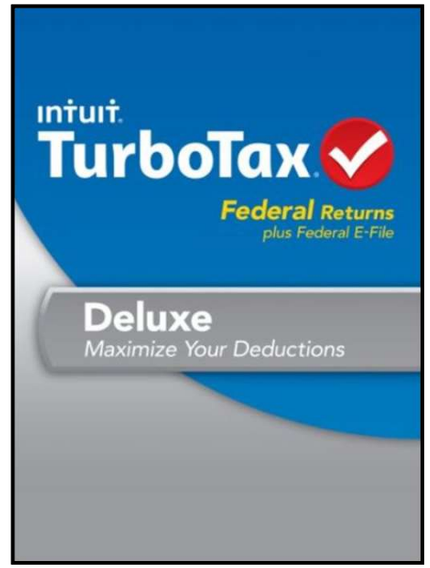
Modular Home

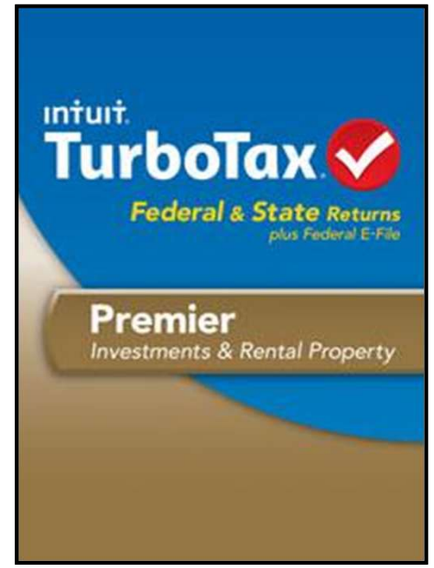
Smartphone OS

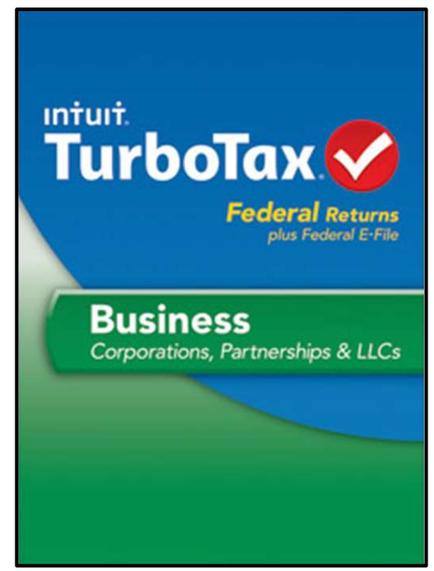
Apple

Modular Product Architecture Example

Modular Software Architecture Example

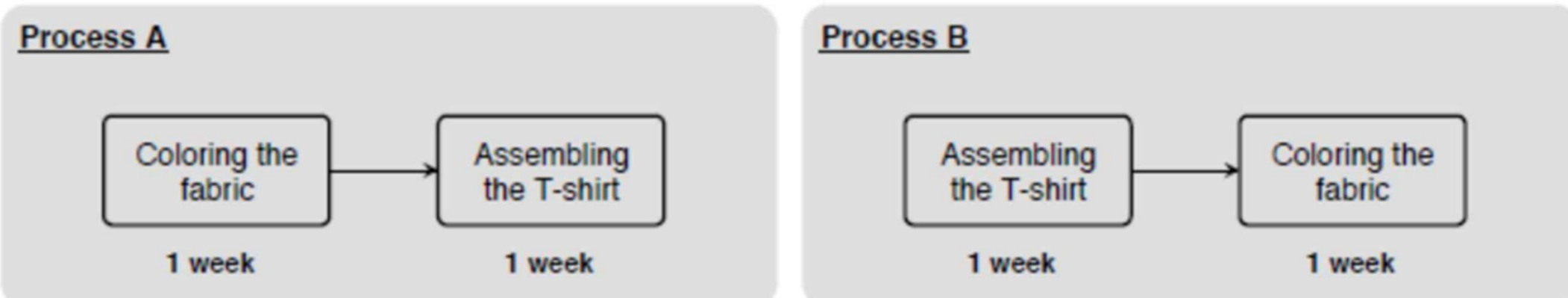






Question:

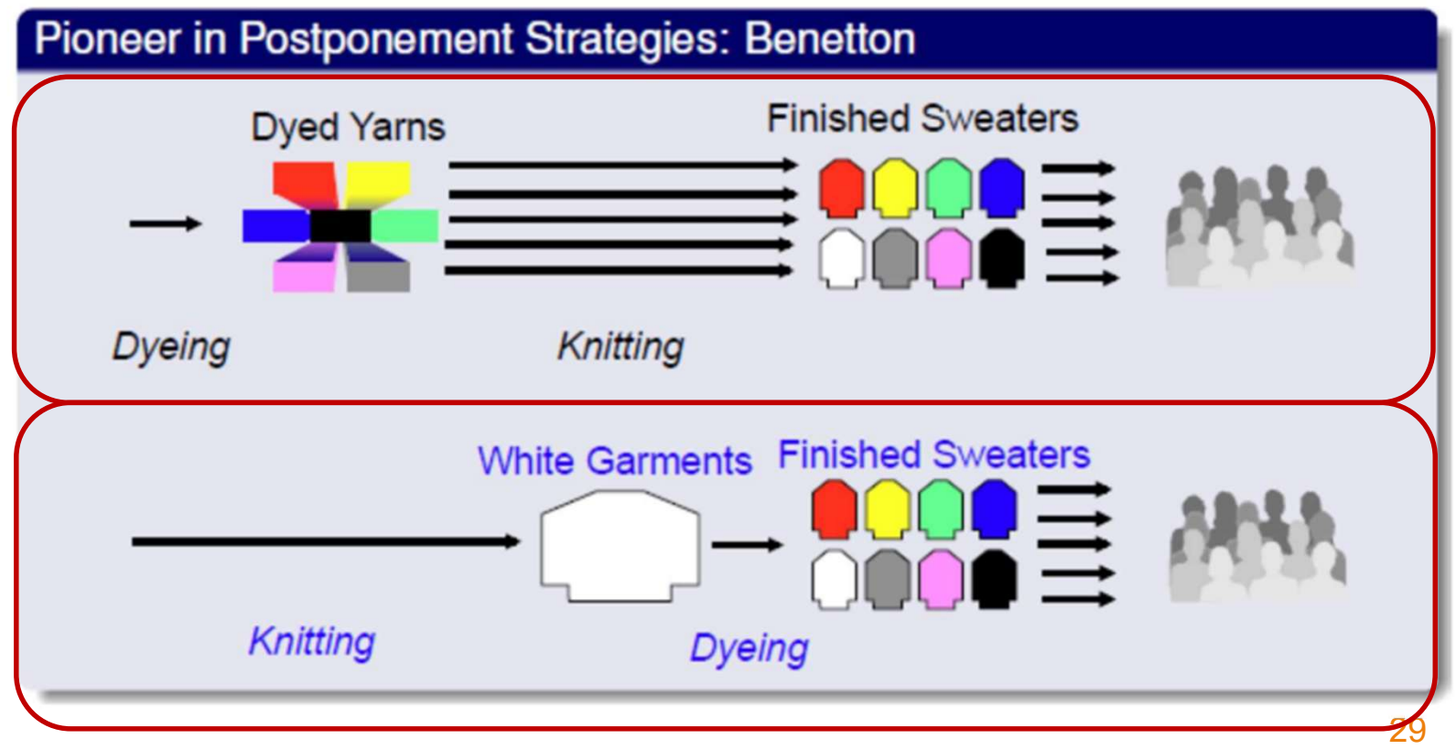
- Suppose you are a garment seller who makes blue, green, purple, and red sweaters. Which one of the two alternative production process will you choose?



Mass Customization Example

- Delayed differentiation (Postponement):
  - The process of producing, but not quite completing, a product or service until customer preferences are known

By using this strategy, Benetton was able to reduce unsold stock from 33% to 8%

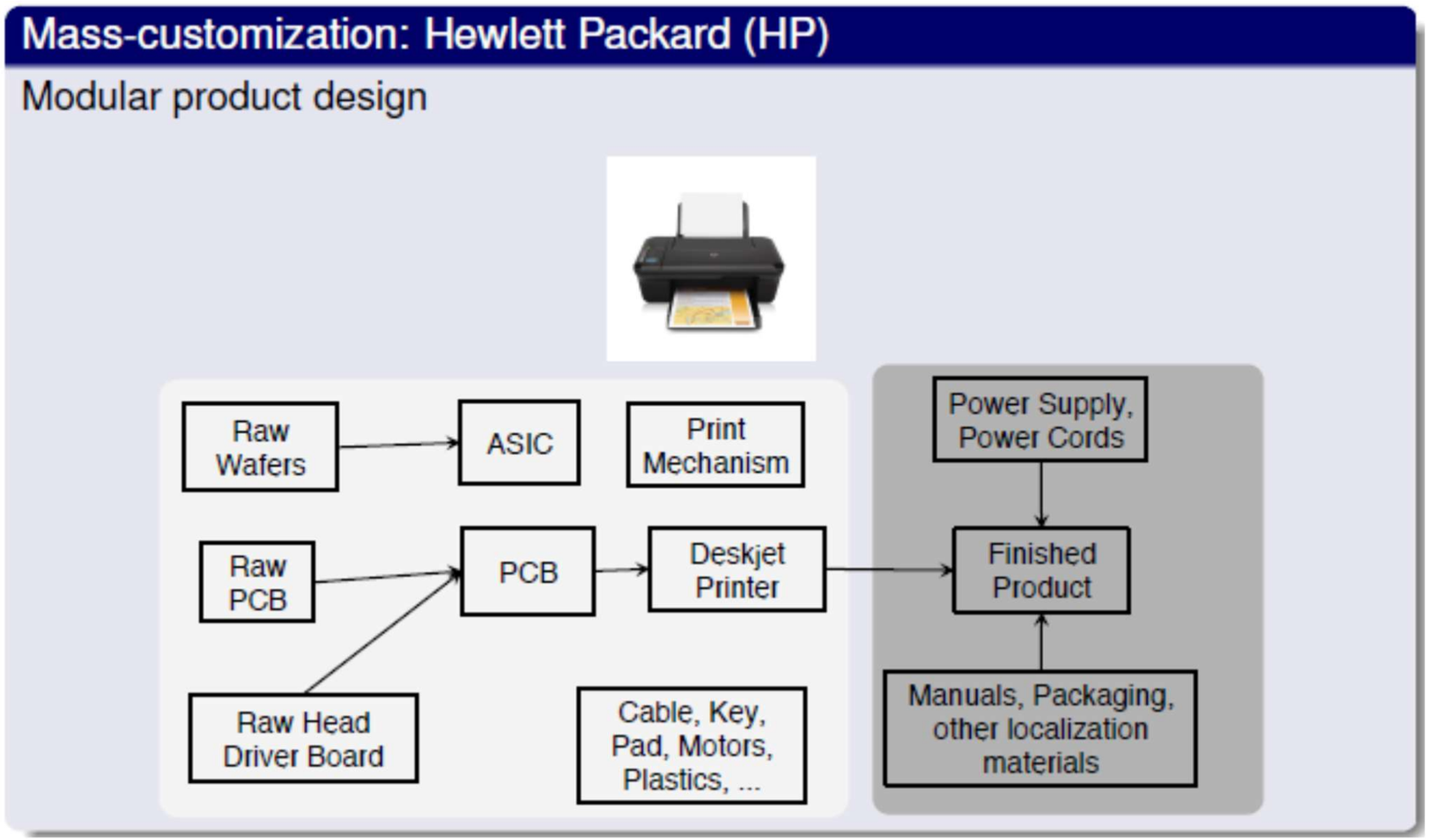


Mass Customization

- Combines mass production process with individual customization
- Read Crawford Textbook Chap 14. p.355

"The key to mass-customizing effectively is postponing the task of differentiating a product for a specific customer until the latest possible point in the supply network."
(Feitzinger and Lee, 1997)

Mass Customization via Modular Design

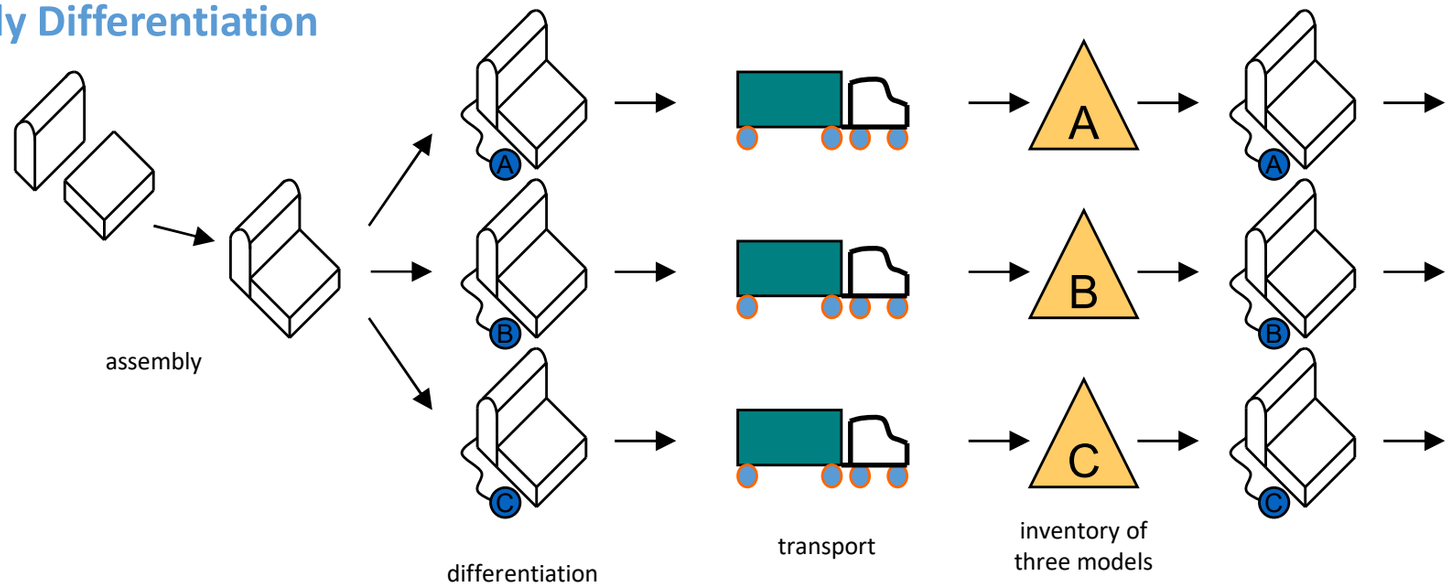


Modular product design:

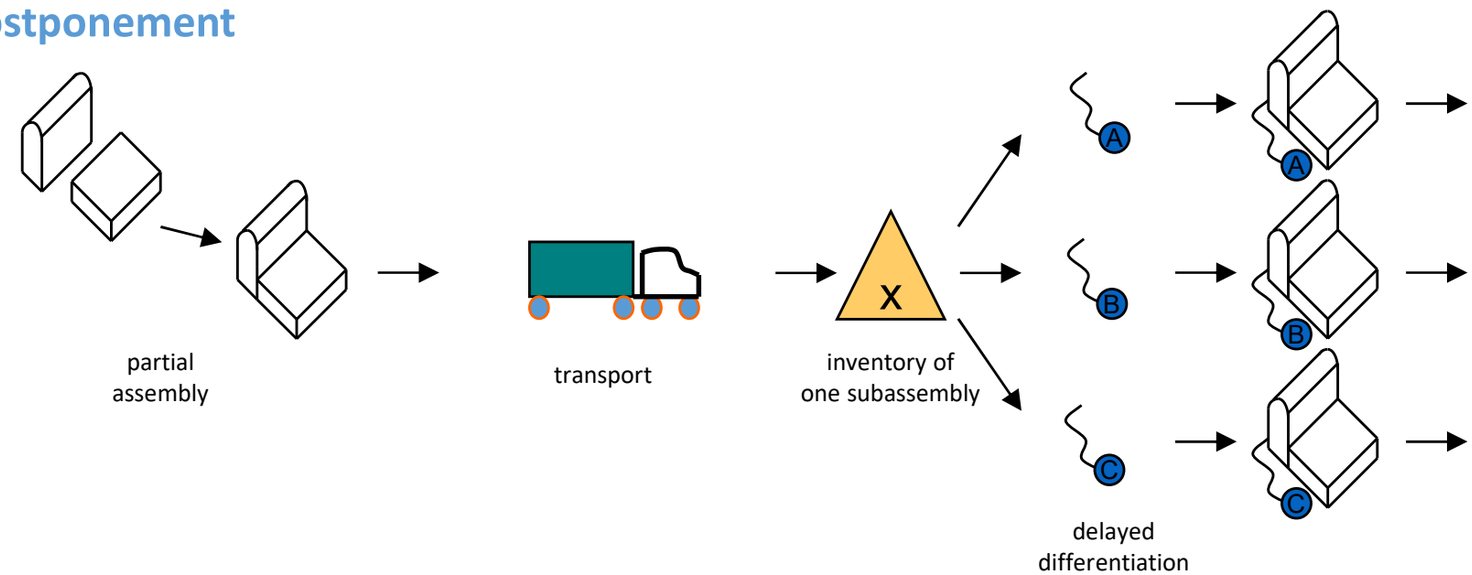
A form of standardization in which component parts are grouped in modules that are easily replaced or interchanged

Modularity Enables Delayed Differentiation

Early Differentiation



Postponement



Benefit of Postponement

- Increased forecasting accuracy
  - Need only aggregate estimates (total number of items regardless of colors)
  - Actual sales data (for immediate selling season) can be used from the test batch

Gillette packaging is postponed in accordance with specific country and retail-chain forecasts

What and when to postpone?

- ✓ Uncertain demand for many product variants
- ✓ Short customer lead time
- ✓ Technologically feasible for **modular** design

Product Design

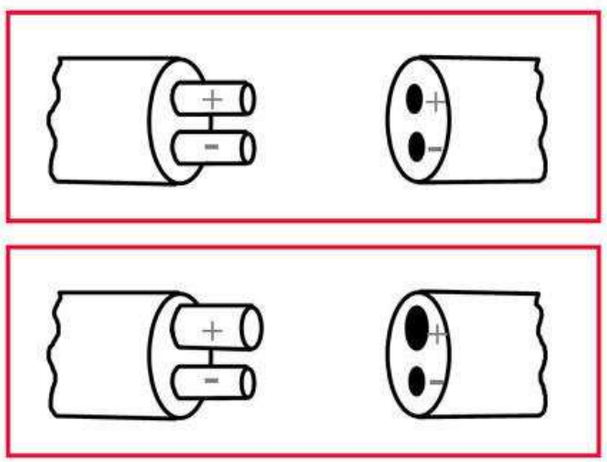
- Product design: The process to blend customer needs with business goal to make successful product

[What is product design? Video]

Fail-Proof Product Design

<u>Poka-Yoke: Fail-Safe method</u>

- Building safeguards into a process to reduce or eliminate the potential for errors during a process.
- Examples: https://youtu.be/K8GQvF004-Y



Product Design and Architecture

- Choice of product design and architecture should reflect what may happen in the market (solve customers' problem!)

- Fail-proof product design
  - Preemptively built-in safeguard design to prevent mistakes or human errors

- Mass customization (Crawford (textbook) Chapter 14 p.354-360)
  - Needs production/inventory strategies aligned with marketing
    - Read: Feitzinger, E., & Lee, H. L. (1997). Mass customization at Hewlett-Packard: the power of postponement. Harvard business review, 75, 116-123.
  - Can reduce the risk of unsold stocks with the right process redesign

Review Today

- Conjoint analysis as concept and market testing tool
- Product development considerations
  - Product architecture and design
- Product launch strategy

Next Class

- Analytic Tools for Market Testing: Designing “Experiments”
  - Motivation: Causality vs. Incentive Compatibility
  - A/B testing
  - Factorial design
  - Prediction market

- Project Management

Group Project Office Hours

- You should have already found the core competence of team members
  - Different skills, experience, knowledge- plan for delegation of responsibility

- Each group will present the concept to me in office hours
  - Think of a new product/service that will meet the needs of your identified target customers

- Your concept should include:
  - A description of customer needs
  - Target customers
  - Brief description of your product or service concept
  - Preliminary view of market potentials